

TUNABLE NON-LINEAR MANIFOLD ROMs FOR EL-LIPTIC, PARABOLIC, AND HYPERBOLIC PDES VIA MATRIX-FREE PETROV–GALERKIN PROJECTION

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ABSTRACT

Learned PDE surrogates such as Fourier Neural Operators and DeepONets typically deliver one (accuracy, speed) operating point per trained model; moving it generally requires retraining. We present a nonlinear-manifold reduced-order model (NM-ROM) for Poisson, heat and viscous Burgers equations that exposes a deployment-time accuracy–speed tradeoff from a single trained decoder, controlled at inference time by the correction rank q (accuracy) and the solver tolerance, budget and empirical-quadrature rule (cost). The framework combines a frozen coordinate-network spatial bank with a small nonlinear head and nested correction directions; exact Dirichlet enforcement by a vanishing factor in the bank; least-squares Petrov–Galerkin projection of the discrete residual onto fixed weak tests, with every linear operator preassembled; matrix-free JAX evaluation; and NNLS empirical quadrature for the Burgers advection, validated on held-out reached states. Across Poisson in two and three dimensions, and heat and viscous Burgers in two, at meshes up to 4096^2 , the same frozen model keeps its accuracy as the two-dimensional mesh is refined, so its speedup over the named iterative solvers grows with resolution, reaching $146\times$ at 0.96% error against conjugate gradients on Poisson, $36.0\times$ at 0.49% against Crank–Nicolson CG on a sealed held-out heat cohort, and $4.87\times$ at 0.60% on Burgers against Newton–BiCGStab ($5.38\times$ at 1.33% on held-out cases); at matched latent dimension its fast setting is far more accurate than a reproduced shallow-masked-autoencoder NM-ROM and POD-LSPG. The named solvers remain more accurate at their tolerances, and three-dimensional Burgers and heat, Navier–Stokes and full-state waves miss their targets.

1 INTRODUCTION

Learned PDE surrogates such as neural operators (Li et al., 2021; Lu et al., 2021; Kovachki et al., 2023) typically deliver one accuracy–speed operating point per trained model; moving it generally requires retraining. Projection-based reduced-order models (ROMs) solve the governing equations in a small trial space (Benner et al., 2015; Sirovich, 1987). A linear subspace is inexpensive to use, but can require many basis functions to represent moving fronts; nonlinear-manifold ROMs learn a more compact representation (Lee and Carlberg, 2020; Kim et al., 2022), although their online nonlinear solve can offset the benefit of fewer unknowns. A deployed ROM usually offers one operating point, whereas a practitioner may want to spend more computation on a difficult query or accept a less accurate answer when response time matters, and the cost of a full-order solve grows with the mesh. We ask: *can one trained NM-ROM give a deployment-time accuracy–cost choice, keep its accuracy as the mesh is refined, and be faster than named iterative full-order solvers while being more accurate than prior nonlinear-manifold ROMs?*

Our model represents the solution by a learned spatial bank multiplied by a nonlinear map of latent variables, augmented with nested, precomputed correction directions: activating more directions enlarges the trial space without changing the learned weights. The correction rank is the representation control; stopping tolerances and empirical quadrature set the cost of the solve. All online coefficients come from the weak PDE residual, without access to the reference solution.

For two-dimensional Poisson, heat and viscous Burgers the answer is largely yes (Table 1, Figure 1): the accurate error stays near its coarse-mesh level up to 4096^2 while the speedup over the named solver grows, and on held-out Burgers cases the model is far more accurate than a reproduced NM-ROM and POD-LSPG (Table 2). It is not yet so for three-dimensional Burgers and heat, Navier–Stokes or the full wave state (Table 3).

Contributions.

1. **A tunable NM-ROM from one trained model.** Nested correction directions provide discrete deployment settings that change approximation capacity without retraining. We isolate correction rank at fixed residual test count and separately evaluate a prescribed test-count schedule across seeds and held-out cases.
2. **An architecture supporting the accuracy–cost tradeoff.** A learned spatial bank, a nonlinear head with a linear skip, and nested corrections support compact latent solves, boundary enforcement and node-local evaluation. The nonlinear coordinates and linear corrections have distinct roles in the reduced representation.
3. **Resolution-robust speedups over named iterative solvers.** With precomputed linear operators, matrix-free differentiation and validated empirical quadrature, the accurate setting reaches 0.96 % on Poisson at $146\times$ the speed of CG, and 0.60 % on Burgers at $4.87\times$ that of Newton–BiCGStab, at 4096^2 in the same allocation.

2 RELATED WORK

Projection-based and nonlinear-manifold ROMs. Linear ROMs approximate solutions in a fixed subspace (Benner et al., 2015; Sirovich, 1987); moving fronts can require a large basis (Cohen and DeVore, 2015; Greif and Urban, 2019). Nonlinear-manifold ROMs combine learned representations with residual projection (Lee and Carlberg, 2020; Kim et al., 2022), including quadratic manifolds (Geelen et al., 2022; Barnett and Farhat, 2022), neural-field representations (Kim et al., 2024) and adaptive bases (Peherstorfer and Willcox, 2015; Carlberg, 2015). Our focus is a nested set of correction directions within one learned bank, selected after training. Least-squares Petrov–Galerkin projection (Carlberg et al., 2011) and preassembly of reduced operators (Ștefănescu and Sandu, 2014; Weder et al., 2024) are established components; the experiment isolates the effect of correction rank.

Neural operators and hyper-reduction. FNO (Li et al., 2021), DeepONet (Lu et al., 2021), U-Net (Ronneberger et al., 2015; Takamoto et al., 2022) and Transolver (Wu et al., 2024) learn solution maps. Their evaluation resolution or inference procedure can also change accuracy and cost; our mechanism instead changes the trial space used to solve the PDE. Empirical quadrature (Hernández et al., 2017; Yano and Patera, 2019) reduces nonlinear residual evaluation through non-negative weights. We retain that construction, testing its error on states reached by the solver and through independent re-draws. The main iterative linear-system comparator is conjugate gradient (Hestenes and Stiefel, 1952).

3 METHODOLOGY

We turn a learned manifold into a practical PDE solver through three components: a trial manifold whose decoder enforces Dirichlet conditions exactly and carries nested linear correction directions (§3.1); a least-squares Petrov–Galerkin projection of the discrete residual onto fixed weak tests (§3.2); and exact preassembly of every linear term with Empirical Quadrature for Burgers advection where validated (§3.3); §3.4 gives the architecture. Throughout, $u \in \mathbb{R}^n$ is the full-order state on a uniform grid of N intervals per axis, $A \in \mathbb{R}^{n \times n}$ the discrete negative Laplacian; R is the bank width, k the latent dimension, M the number of weak tests and m the number of quadrature nodes. The formulas below illustrate scalar Dirichlet problems; periodic vector fields and time-integrator details are specified per PDE in the appendix. Figure B.1 (Appendix B) shows the data flow; Appendix A gives the per-PDE derivations and exit codes.

3.1 TRIAL MANIFOLD WITH EXACT DIRICHLET ENFORCEMENT

We train a decoder $\mathcal{D} : \mathbb{R}^k \rightarrow \mathbb{R}^n$, $k \ll n$, with latent state z and no encoder in the deployed path, separable into a frozen spatial *bank* $G \in \mathbb{R}^{n \times R}$ times a small neural *head* $h_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^R$, $k \leq R \ll n$, the bank built once per mesh from a random-Fourier-feature coordinate network g_ϕ (Tancik et al., 2020),

$$G_{x,:} = \mu(x) g_\phi(x)^\top, \quad \mu(x) = 16 x_1(1 - x_1) x_2(1 - x_2). \quad (1)$$

On the two-dimensional square the smooth vanishing factor μ of (1), zero on the boundary Γ , is folded into every column of the bank, so $\partial u / \partial z = 0$ on Γ at every resolution (homogeneous data only); the L-shape uses a distance-based factor, the cube a scaled product factor, and periodic Navier–Stokes none (§3.4).

Nested correction directions. The deployed model augments the head’s output with q fixed directions $C_q \in \mathbb{R}^{R \times q}$ and solves for the latent code and the correction coefficients together,

$$u(z, y) = G(h_\theta(z) + C_q y), \quad z \in \mathbb{R}^k, y \in \mathbb{R}^q, 0 \leq q \leq R. \quad (2)$$

The columns of C_q are chosen offline from training data; the Burgers construction uses principal components of the head’s coefficient errors. Per-PDE constructions are stated in Appendix A. The directions are ordered so that C_q is a prefix of $C_{q'}$ for $q < q'$; the ladder is nested and no rung needs retraining. At $q = 0$ the model is the frozen head; at $q = R$ the head is irrelevant and the representation is the full linear span of the bank. For linear PDEs its coefficients can be found by a reduced linear solve; for nonlinear PDEs the equations can remain nonlinear even in this full-bank representation; in between, q moves the reachable set from the head’s image towards the bank’s span, and the solved dimension from k to $k + q$. Two consequences follow: the bank’s projection error is a floor on the achievable error whatever the head or the solver does, so nonlinearity in h_θ buys a smaller *solved* dimension, not an escape from the span; and all x -dependence factors through G , which makes node sampling and exact preassembly available from one decoder.

3.2 PROJECTING THE PDE ONTO THE MANIFOLD

Each PDE we consider gives a discretised residual $r(u)$ that should vanish at the true solution. We substitute the trial manifold $u(z, y)$ into r and project onto a fixed, latent-independent test space. On the Dirichlet square, $P \in \mathbb{R}^{M \times n}$ collects low-frequency tensor-product sine vectors, eigenvectors of the five-point operator, $PA = \Lambda P$; the L-shape uses Lanczos eigenvectors of its own operator. The reduced problem solved online is

$$(z^*, y^*) = \arg \min_{z, y} \frac{1}{2} \|r_w(z, y)\|_2^2, \quad r_w(z, y) = \Lambda_\star^{-1} P r(u(z, y)) \in \mathbb{R}^M, \quad M > k + q, \quad (3)$$

with a diagonal row scaling $\Lambda_\star$ stated per PDE in Appendix A: a least-squares Petrov–Galerkin condition with an explicit test space (Carlberg et al., 2011; Lee and Carlberg, 2020), not tangent Galerkin, overdetermined rather than square, solved by a method appropriate to each PDE’s structure.

Elliptic (Poisson). The Poisson residual is $r(u) = Au - f$, and projection gives $r_w(z, y) = B_0(h_\theta(z) + C_q y) - b_0$, $B_0 = PG$ (Appendix A.1); the projector onto $\text{range}(B_0 C_q)$ does not depend on z , so y is eliminated exactly (Golub and Pereyra, 1973), the iteration stays k -dimensional at every q , and $q = R$ is a linear solve.

Parabolic (heat). The heat semi-discretisation $du/dt = -\kappa Au$ is advanced with Crank–Nicolson; the reduced step substitutes the manifold into the fully discrete equation *before* projecting and solves $z_{n+1} = \arg \min_z \|B_0 h_\theta(z) - D B_0 h_\theta(z_n)\|_2$, with D the diagonal Crank–Nicolson amplification of the tests (Appendix eq. (7)) (shown at $q = 0$; with corrections see Appendix A.2), a nonlinear least-squares problem, not a linear system; at $q = R$ the step is the linear recurrence (Appendix eq. (9)). Reflective waves use an explicit latent integration instead (Appendix A.4).

Hyperbolic (Burgers). For $u_t + u(u_x + u_y) = \nu \Delta u$ with a sign-upwind stencil and backward Euler (Appendix A.3) the weak residual is nonlinear in the coefficients, so y is not eliminated in closed form; we damp the (z, y) blocks separately inside one Levenberg–Marquardt step (*block-damped* variable projection), which, in the recorded solver comparison, removes the joint LM configuration’s 6 budget exits (Table D.3).

Solver and exits. Each attempt solves the damped normal system $(H + \lambda \text{diag}(\text{diag } H)) \delta = -g$, $H = J^\top J$, $g = J^\top r_w$, accepting the step only if it strictly decreases the residual: damped Levenberg–Marquardt with a monotone test (Marquardt, 1963). Three exit families are recorded and never conflated: the scale-free stationarity measure $\eta(z, y) = \|J^\top r_w\|_2 / (\|J\|_F \|r_w\|_2) \leq \eta_{\text{tol}}$ (4), a residual threshold, and the stalls, reported as early-stopped, never as converged. No query uses the solution it predicts: the elliptic solve starts from the cached training code nearest the projected source; the heat and Burgers queries start from nearest codes and fit (z, y) to the supplied initial field (Appendix A.6).

3.3 HYPER-REDUCTION

The non-linear manifold reduces the DOF count from n to $k + q$, but a projected term such as $PN(Gc)$ still costs $O(n)$. For a fixed linear operator the tested residual is exactly precomputable, $P(Au - f) = B(h_\theta(z) + C_q y) - b$ with $B = \Lambda PG = \Lambda B_0$ (the B_0 of §3.2 with its rows rescaled) assembled offline by the corresponding discrete transforms, so linear terms are never approximated; the Burgers advection $PN(Gc)$ is the only term that resists preassembly, evaluated *densely*, $O(nR)$ per residual, or on an *empirical quadrature* rule of m nodes (Hernández et al., 2017; Yano and Patera, 2019), $O(mR)$. The rule is a non-negative *per-node* weight vector, *independent of the snapshot*, fitted by NNLS (Lawson and Hanson, 1974) to reproduce the projected advection term at n_{fit} stored codes with a hard cap of m nodes (Appendix A.5); changing m re-solves the fit, rules are not nested, and deployment selects among stored rules. *A rule is never accepted on its NNLS fit residual*; we score it by the held-out relative error ρ of the projected advection term (Appendix eq. (14)) over states the solver actually reaches on trajectories disjoint from the fit and evaluation cases, against a primary bar $\rho_{\text{max}} \leq 0.116$ fixed before any certification job ran and a tight bar 0.06, and then re-draw the construction: a rule is *confirmed* only if every re-draw passes (Table D.2).

3.4 MODEL ARCHITECTURE

Every query starts from stored training codes, never cold: the elliptic solve from the code nearest the projected source, the time-dependent solves from nearest codes refined by a fit to the supplied initial field, then warm-started from step to step. No encoder is used.

Head. $h_\theta(z) = \varphi_\theta(z) + W^\top z$, with φ_θ an MLP with two hidden SiLU layers; the skip keeps a latent-independent Jacobian component. Widths and (k, R) are per family (Table C.2; three-dimensional sizes in Table C.11). The skip is a design choice, not ablated here.

Bank. Every Dirichlet bank is a coordinate network times a vanishing factor: μ of (1) on the square, its product form in 3D, a distance-based factor on the L-shape; some banks are right-multiplied by a fixed orthonormalising matrix. The network’s parameters do not depend on the mesh, so a quadrature rule’s support is a cached block, and decoding in row blocks at large n changes memory, not the model. The periodic Navier–Stokes bank has no factor and a global solenoidal projection.

Corrections. C_q holds leading singular directions of training coefficient residuals, nested in q .

What is fixed, what is chosen, and how error is reported. Each row of Table 1 comes from one frozen model, named by its problem label. Chosen at run time are q , the number of tests M (usually $M = 4(k + q)$; fixed along the rank study of Table C.8 and on the L-shape), the quadrature, the tolerance and budget, and for heat the stepping mode (Crank–Nicolson or the batched exact-propagator fit). The representation ablations distinguish three quantities: the *bank floor* (projection error onto range G), the *best-found* error (the smallest any point of the augmented manifold attains, from a multistart oracle) and the *solved* error the iteration returns.

4 IMPLEMENTATION

4.1 MATRIX-FREE PROJECTED OPERATORS VIA JAX

Every linear term is preassembled once per mesh as the $M \times R$ matrix B , so the online residual and Jacobian of a linear PDE are $B(h_\theta(z) + C_q y) - b$ and $B[Dh_\theta \ C_q]$, with Dh_θ from a forward-mode

Jacobian-Vector Product (Bradbury et al., 2018); the sampled Burgers advection uses the cached stencil block. The damped normal system is solved directly; no Krylov solve is applied to the projected operator. The Burgers runs above 1024^2 assemble the Jacobian analytically instead of by forward-mode products, clip the step, and (in the settings listed in Table C.4) solve the normal system by Cholesky factorisation, carry the damping between time steps and start each step from a quadratic predictor.

4.2 TRAINING PROTOCOL

The bank and the head are trained in two stages, both as auto-decoders: the coordinate network g_ϕ is fitted to the training states through (1), then frozen, and the head is fitted with a code library Z on the bank-projected states; C_q is then constructed from training-only coefficient directions as specified for each PDE. Sizes and cohorts are in Appendix C; checkpoint hashes and recorded offline costs are retained with the source evidence described there.

5 EXPERIMENTAL SETUP AND BENCHMARK PROBLEMS

Problems and references. The two-dimensional problems are Poisson on the square and on an L-shaped domain, heat, viscous Burgers and reflective waves; the three-dimensional problems are Poisson, heat, Burgers and incompressible Navier–Stokes. Table C.2 gives the two-dimensional meshes and cohorts, Table C.3 the sampled families and Table C.11 the three-dimensional configurations. Burgers provides the correction-rank study.

Accuracy. We report worst relative L^2 error over the stated cases and requested times, in percent. Poisson uses the steady reference norm and heat the current reference norm; Burgers and Navier–Stokes use the initial-field norm over evolved times; waves use the energy-state error over displacement and velocity. Same-grid error measures reduction error relative to the converged discrete solution. Error against a refined reference also includes discretisation error; the two are never interchanged. Settings for a held-out (final) cohort are fixed before its cases are accessed; all other cohorts are development cohorts.

Timing and speedup. Every accuracy–runtime pair comes from the same solver invocation. We use double precision, GPU burn-in, synchronised timing and medians of retained repetitions. GPU query time includes initialization, the solve or rollout, and requested device outputs. Complete-query time additionally includes host transfers and is labelled separately. Training and compilation are offline costs. Speedup is $S = T_{\text{FOM}}/T_{\text{method}}$ with both times from the same allocation; each table names the FOM algorithm and shows its error. Solves that miss their stopping rule are reported as such and do not enter a speedup.

Baselines. Linear problems are compared with conjugate gradients (CG), including the L-shaped domain and the CG solve inside each Crank–Nicolson (CN) heat step. Burgers is compared with Newton–BiCGStab and Navier–Stokes with its CNAB2 time integrator. The named FOM of a row is that solver at its fastest tested setting that is at least as accurate as the accurate NM-ROM setting.

6 NUMERICAL EXPERIMENTS AND RESULTS

Table 1 is the headline comparison; the subsections that follow give the settings behind it and the controls that move a frozen model between them.

6.1 ACCURACY AND SPEED AGAINST FULL-ORDER SOLVERS

Linear problems. On Poisson the accurate setting reaches 0.96 % error and is $13.9\times$ faster than CG at 1024^2 ; the corrections cost little speed because they are eliminated analytically (§3.2). The error stays at 0.96 % up to 4096^2 while the GPU-query speedup rises to $146\times$. On the L-shaped domain the accurate setting is $6.07\times$ faster at 2.12 %. The earlier heat checkpoint crosses CG between 256^2 and 1024^2 ($4.85\times$ at 1024^2). A wider heat bank, evaluated once on a sealed held-out cohort at

Table 1: NM-ROM against the named full-order solver at every measured problem and mesh. **Both NM-ROM columns of a row come from one frozen model:** *fast* is the uncorrected setting ($q = 0$) and *accurate* the selected accurate rank (Table C.4); no network is retrained between them. Error is the worst relative L^2 error over the cohort (%). Speedup is the FOM’s median time divided by the NM-ROM’s in the same GPU allocation, **bold** where the NM-ROM is faster. The FOM is the named iterative solver at its fastest tested setting that is at least as accurate as the accurate setting. ^f held-out final cohort (all others development); ^r error against a refined reference, which includes discretisation error (all others same-grid); ^t all output times including $t = 0$; ^e evolved output times only; ^c complete-query time with host transfers (all others GPU query; heat was measured in GPU query only). Heat (wide bank) is a separately trained frozen model with a 128-function bank ($q = 32$ accurate); over evolved times its accurate error is 0.22 % (0.21 % batched). The batched fit solves each output time independently against exactly propagated test moments, exploiting the linear, autonomous structure of the heat equation (eigenfunction tests). Burgers accurate quadrature: ^s stored rule of job 3780164 that passed the held-out bar in its single draw and was not re-drawn (it is not the $q = 256$ rule of Table D.2); ^ℓ deterministic 63×63 lattice rule passing the held-out bar in the same job; ^d dense residual. The earlier Burgers model has one setting, may exit on a stall, and is shown against relaxed ($10^{-2}/0.5$) and tight ($10^{-6}/10^{-8}$) nonlinear/linear Newton tolerances (other rows’ tight-Newton ratios: Table C.4). ^h 64 held-out cases never used for selection (not the sealed final cohort); at 2048^2 the development-chosen $M = 544$ setting was not run on them, so the $M = 1088$ setting is shown. Poisson (dev. sources) rows use development sources of a separate run and are not the held-out cohort. Complete-query speedups: Table C.4. Dashes mark settings not measured. Times and settings: Table C.4.

Problem	Mesh	NM-ROM accurate		NM-ROM fast		Full-order model (FOM)	
		Err. (%)	Speedup	Err. (%)	Speedup	Err. (%)	Solver
<i>Two-dimensional</i>							
Poisson	256 ²	0.97	3.13 ×	3.16	3.33 ×	0.15	CG, rtol 10 ^{−2}
	1024 ²	0.96	13.9 ×	3.15	15.2 ×	0.072	CG, rtol 10 ^{−2}
	2048 ²	0.96	72.8 ×	3.15	74.3 ×	0.049	CG, rtol 10 ^{−2}
	4096 ²	0.96	146 ×	3.15	148 ×	0.038	CG, rtol 10 ^{−2}
Poisson, L-shape ^c	256 ²	2.13	3.84 ×	3.86	4.04 ×	0.64	CG, rtol 10 ^{−2}
	512 ²	2.12	6.07 ×	3.85	6.18 ×	0.38	CG, rtol 10 ^{−2}
	1024 ²	2.20	9.76 ×	3.85	10.0 ×	0.31	CG, rtol 10 ^{−2}
	2048 ²	2.20	19.3 ×	3.85	19.4 ×	0.24	CG, rtol 10 ^{−2}
Heat ^r	64 ²	—	—	4.56	0.22×	1.59	CN-CG, rtol 10 ^{−2}
	256 ²	—	—	4.56	0.61×	0.93	CN-CG, rtol 10 ^{−2}
	1024 ²	—	—	4.56	4.85 ×	0.77	CN-CG, rtol 10 ^{−2}
Heat (wide bank) ^{t,f}	1024 ²	0.49	1.59 ×	1.36	1.57 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	2048 ²	0.49	9.74 ×	1.36	9.59 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	4096 ²	0.49	36.0 ×	1.36	35.4 ×	0.47	CN-CG, Δt=0.05, rtol 10 ^{−2}
Heat (wide bank, batched fit) ^{t,f}	1024 ²	0.49	13.9 ×	1.36	13.0 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	2048 ²	0.49	60.0 ×	1.36	57.0 ×	0.18	CN-CG, Δt=0.05, rtol 10 ^{−3}
	4096 ²	0.49	103 ×	1.36	101 ×	0.47	CN-CG, Δt=0.05, rtol 10 ^{−2}
Burgers ^e	256 ²	0.51 ^s	0.043×	1.89	0.79×	0.049	Newton-BiCGStab, tol 10 ^{−3}
	512 ²	0.55 ^s	0.068×	2.14	1.31 ×	0.052	Newton-BiCGStab, tol 10 ^{−3}
	1024 ²	0.59 ^d	0.0037×	2.29	2.02 ×	0.034	Newton-BiCGStab, tol 10 ^{−4}
Burgers ^e	2048 ²	0.87 ^ℓ	0.99×	2.37	4.16 ×	0.049	Newton-BiCGStab, tol 3×10 ^{−3}
	4096 ²	0.60 ^ℓ	4.87 ×	2.41	12.9 ×	0.050	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers (held-out cases) ^{e,h}	2048 ²	1.31 ^ℓ	1.43 ×	9.03	5.63 ×	0.13	Newton-BiCGStab, tol 10 ^{−3}
	4096 ²	1.33 ^ℓ	5.38 ×	9.03	13.2 ×	0.14	Newton-BiCGStab, tol 3×10 ^{−3}
Burgers (earlier model) ^r	1024 ²	—	—	3.91	1.63 ×	2.39	Newton-BiCGStab, relaxed
	1024 ²	—	—	3.91	14.5 ×	2.14	Newton-BiCGStab, tight
Burgers, certified quadrature rule	256 ² , 512 ² , 1024 ²	<i>results incoming</i>					
Burgers (wider learned bank), held-out 64	2048 ² , 4096 ²	<i>results incoming</i>					
<i>Three-dimensional</i>							
Poisson ^f	32 ³	0.26	0.94×	1.40	0.99×	0.16	CG, rtol 10 ^{−2}
	64 ³	0.26	1.33 ×	1.39	1.38 ×	0.11	CG, rtol 10 ^{−2}
Poisson (dev. sources)	128 ³	0.16	6.75 ×	0.55	6.98 ×	0.075	CG, rtol 10 ^{−2}
	256 ³	0.16	23.2 ×	0.55	23.7 ×	0.049	CG, rtol 10 ^{−2}
Heat 3D (wider bank)	64 ³ , 128 ³	<i>results incoming</i>					

every mesh, reaches 0.49 % over all times and is $36.0 \times$ faster than CN-CG at 4096^2 ($103 \times$ with the batched fit). In three dimensions the held-out Poisson cohort gives 0.26 % at $1.33 \times$.

Burgers. The reduced solve costs the same at every mesh, so the fast setting passes Newton-BiCGStab at 512^2 and is $2.02 \times$ faster at 1024^2 with 2.29 % error. The accurate setting reaches 0.51 % at 256^2 and, on development cases, is slower than the FOM through 2048^2 . At 4096^2 the accurate setting chosen on six development cases reaches 0.60 % and is $4.87 \times$ faster than the fastest Newton setting at least as accurate; on 64 held-out cases the same setting gives 1.33 % at $5.38 \times$ (audit note in Appendix C.1). A certified quadrature rule at 256^2 – 1024^2 and a wider, held-out-tested bank are in progress (results incoming).

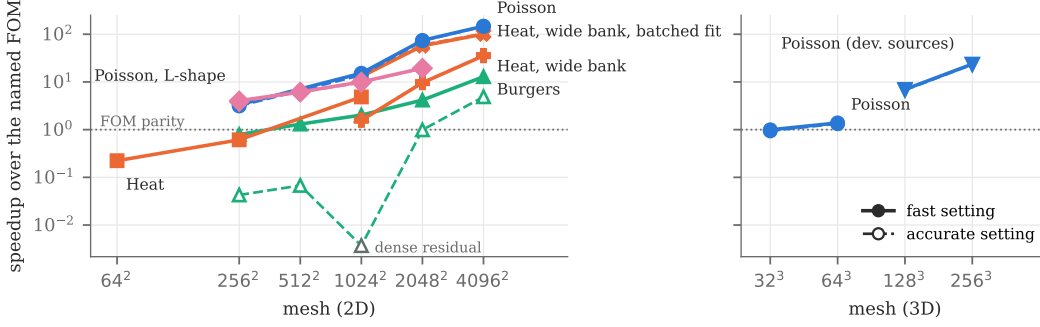


Figure 1: Speedup over the named FOM against mesh, plotted from the rows of Table 1. The grey accurate Burgers point at 1024^2 is dense (no validated $q = 256$ rule there; the validated $q = 128$ rule gives 1.04% at $0.50\times$). Burgers points at 2048^2 and 4096^2 are development cases; the held-out rows of Table 1 are not plotted. The legend applies to both panels.

Table 2: Other nonlinear-manifold ROMs on the shared Burgers 2D family at 256^2 and 512^2 : worst and median evolved same-grid error over 32 validation cases (never used for any choice) and compiled-query memory, one allocation. Latent dimension is matched for the baselines and our fast setting; our accurate setting solves $k + q = 272$ unknowns. The shallow masked-autoencoder NM-LSPG of Kim et al. (2022) passed its reproduction gate, on their own 2D Burgers benchmark (their Sec. 6.2), on the third of three pre-registered attempts (sigmoid activation; median 1.45% against a 1.5% bar; published $<1\%$); its hyper-reduction was not reproduced, and its published encoder width ($2n$) needs 136 GB to train at 256^2 and 2183 GB at 512^2 , beyond the 77 and 135 GB devices; at 512^2 its encoder, capped at width 4096, trained only 229–231 epochs in its 4800 s budget, so those errors partly reflect a training-time limit. Data-matched: trained on our bank’s 576 trajectories. The convolutional-autoencoder NM-ROM (Lee and Carlberg, 2020) was not run. No times: every row, ours included, uses a dense residual (Tables 1, C.7).

Method	Unknowns	256 ²			512 ²		
		Worst (%)	Median (%)	MB	Worst (%)	Median (%)	MB
Kim et al. NM-LSPG	8	163.93	43.34	2480	175.71	55.48	9959
Kim et al. NM-LSPG	16	145.40	43.87	2567	174.99	58.56	10511
Kim et al. NM-LSPG	32	127.51	43.61	2742	252.33	56.72	10879
Kim et al. NM-LSPG, data-matched	16	120.33	38.71	2567	—	—	—
POD-LSPG	8	55.52	22.72	18	55.82	24.08	71
POD-LSPG	16	41.53	10.45	30	42.01	13.83	122
POD-LSPG	32	24.78	5.65	55	25.63	7.74	222
This work, fast ($q = 0, k = 16$)	16	6.79	0.66	625	7.72	0.69	2368
This work, accurate ($q = 256, k = 16$)	272	0.88	0.089	1769	1.05	0.080	6891

Resolution. Figure 1 shows that the NM-ROM query time grows more slowly with the mesh than the iterative FOM’s: in every series the fast-setting speedup rises with resolution against the FOM setting selected per row (fastest tested setting at least as accurate as the accurate NM-ROM), fixed across meshes for the CG series and re-selected per mesh for Burgers and wide-bank heat, whose ratios against one fixed tight setting (Table C.4) also rise from 2048^2 to 4096^2 .

On 32 validation Burgers cases at 256^2 and 512^2 (Table 2) our fast setting, at matched latent dimension, reaches worst errors of 6.79–7.72 %, against 120–252 % for the reproduced shallow masked-autoencoder NM-ROM and 25–56 % for POD-LSPG; the accurate setting, at 272 solved unknowns, reaches 0.88–1.05 %.

Where the method currently fails. Table 3 collects the four problems that miss their targets; they are excluded from Table 1. On three-dimensional Burgers the corrections cut the held-out worst error from 16.26 % to 4.40 %, but the FOM is faster, and the refined-reference check fails, so only same-grid reduction error is established. On three-dimensional Navier–Stokes the corrections lower the worst error only from 20.34 % to 18.92 %; 31 of 32 held-out cases miss the 5 % target. On reflective waves the fast setting is faster than CG but its energy-state error, which includes velocity, is 11.34 %; corrections bring it to 5.12 %, still above CG’s, at a cost that removes the speedup. Three-dimensional heat reaches 0.75 % over evolved times ($32^3/64^3$ records hold no other convention) but 1.93 % with

Table 3: Problems on which the method misses its target: Navier–Stokes 5 % per held-out case; heat 1 % over all output times; wave, an energy-state error within that of CG; Burgers 3D, faster than the FOM at its accuracy (no numeric error target was recorded). Error is the worst same-grid relative L^2 error (%) over evolved times, except the Heat 3D 128³ rows (all output times including $t = 0$); the wave error is the energy-state error over displacement and velocity. Speedup is against the named FOM in the same allocation.

Problem	Setting	Err. (%)	FOM err. (%)	Speedup	FOM
Burgers 3D, 32 ³ (final)	$q = 0$	16.26	2.31	0.038×	Newton–BiCGStab
	$q = 192$	4.40	2.31	0.023×	Newton–BiCGStab
Navier–Stokes 3D, 32 ³ (final)	$q = 0$	20.34	2.47	0.0025×	CNAB2
	$q = 256$	18.92	2.47	0.0007×	CNAB2
Wave 2D, 1024 ² (dev.)	$q = 0$	11.34	4.29	3.03 ×	midpoint–CG
	$q = 32$	5.12	4.29	0.23×	midpoint–CG
Heat 3D, 32 ³ (final, evolved)	$q = 0$	1.56	0.32	0.13×	CN–CG
	$q = 96$	0.76	0.32	0.066×	CN–CG
Heat 3D, 64 ³ (final, evolved)	$q = 0$	1.55	0.34	0.27×	CN–CG
	$q = 96$	0.75	0.34	0.13×	CN–CG
Heat 3D, 128 ³ (final, all times)	$q = 0$	3.18	1.58	0.43×	CN–CG
	$q = 96$	1.93	1.58	0.20×	CN–CG

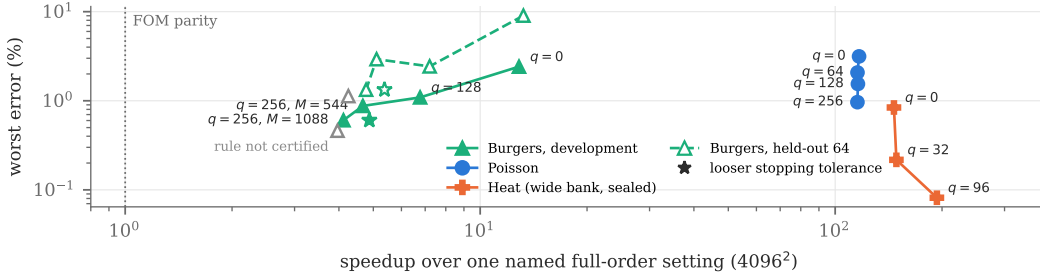


Figure 2: Worst error (evolved times for Burgers and heat) against speed at 4096²; one frozen model and allocation per series, each divided by one full-order setting, the fastest at least as accurate as its best point (Newton–BiCGStab tolerance 3×10^{-3} ; CN–CG $\Delta t = 0.025$, $\text{rtol } 10^{-6}$; CG $\text{rtol } 10^{-1}$, looser than in Table 1). On Burgers q and M change together ($M \approx 4(k + q)$, $M = 544$ an alternative at $q = 256$; fixed- M ladder: Table C.8). Star: looser stopping tolerance; grey: rule failed its held-out check.

$t = 0$ at 128³; it is slower than CN–CG with Crank–Nicolson stepping up to 128³ and faster at 256³ (16 cases) and with the batched fit (Table C.6), but misses the all-times target. The three-dimensional solves evaluate the residual densely. A wider three-dimensional heat bank is in progress (results incoming).

Settings. Table C.4 lists both settings of every row of Table 1 with their times; moving between them changes only deployment arguments. Appendix Table C.8 is the fixed-test-count evidence at 256²: with M held fixed, each added block of correction directions lowers the error and raises the cost.

6.2 WHICH KNOB TO TURN

Appendix Table C.9 summarises the deployment-time controls. Correction rank is the accuracy control (with a smaller fixed test space the same ranks give a 1.22× error reduction from $q = 0$ to 256, so test count is held fixed along the ladder). Empirical quadrature and the stopping tolerance are cost controls: they lower the runtime at little or no change in error (Appendix Table C.10). The iteration cap is a safeguard, not a control; a truncated solve fails.

On Burgers at 4096² the correction rank is a genuine accuracy–cost tradeoff, and every tested setting is faster than the named solver (4.12–12.9× on development cases, Figure 2); on the linear problems the corrections are eliminated analytically, so accuracy improves at nearly constant cost.

A separate ladder with $M = 4(k + q)$, evaluated once on a sealed cohort, gives monotone error reduction for all 4 checkpoints (top-rank errors 0.59–0.68 %, Table D.1); 3 of 4 meet the pre-registered secondary criterion (settings spanning at least $2\times$ in error and in cost). Two stricter pre-registered checks fail: the original model’s sealed-to-development ratio at $q = 0$ (one sealed case converges to a wrong branch, 10.1120 %) and universal convergence (the second retrained seed at $q = 64$ (3 budget exits); Table D.1).

Limitations.

- Speedups are measured against the named iterative full-order solvers at the stated tolerances; comparison with other full-order solver classes (direct and spectral solvers, coarser discretisations) is outside the scope of this study. The named FOM is more accurate than the NM-ROM in every row, and on heat at 4096^2 the linear solve in the learned bank, a baseline (0.13 % in 5.1–8.0 ms), is faster than every NM-ROM setting.
- Accuracy is bounded by the frozen bank: Burgers development accuracy does not carry to held-out cases (the free solve in the same 512-function bank reaches only 1.70 % there); the Poisson errors sit near the bank floor (0.742 % square, 0.143 % cube); the L-shape (2.20 % against a floor of 0.77 %) is limited by the head; the three-dimensional heat bank represents the initial field only to 1.2–1.9 %.
- The fixed-test-space rank study uses one checkpoint; the multi-seed and sealed study sets $M = 4(k + q)$, so it does not replicate it. Comparisons with the FOM do not isolate the nonlinear head’s contribution over the linear span of the same bank.
- Quadrature rules are validated on a finite set of reached states, not certified globally, and none is used in three dimensions; the reduced solve can converge to a wrong branch, and lower same-grid error does not remove discretisation error.
- Except where marked final or held-out, cohorts are small development cohorts, and timings are medians without dispersion.

7 CONCLUSION AND FUTURE WORK

One frozen NM-ROM, with nested corrections that change its approximation capacity after training, keeps its accuracy under mesh refinement on two-dimensional Poisson, heat and Burgers, so its speedup over the named iterative solvers grows with resolution: at 4096^2 its accurate setting reaches 0.96 % at $146\times$ on Poisson, 0.49 % at $36.0\times$ on held-out heat and 1.33 % at $5.38\times$ on held-out Burgers, far more accurate than a reproduced NM-ROM and POD-LSPG. The named solvers remain more accurate, and three-dimensional Burgers, heat and Navier–Stokes and the full wave state miss their targets (Table 3); future work is a better three-dimensional bank, a cheaper high-rank solve and quadrature in three dimensions.

REPRODUCIBILITY STATEMENT

Every table and prose number is generated from hash-pinned run records that identify source, checkpoint, settings, cohort, allocation and audit (Appendix C); failed settings are retained.

AI USE STATEMENT

Language-model assistants supported method and experiment development, implementation, result analysis, auditing, and manuscript drafting and editing. Numerical results were computed by the recorded solver runs; table generators reproduce the reported values from those records. The authors are responsible for verifying the methods, results and text.

REFERENCES

- J. Barnett and C. Farhat. Quadratic approximation manifold for mitigating the Kolmogorov barrier in nonlinear projection-based model order reduction. *Journal of Computational Physics*, 464:111348, 2022.

- P. Benner, S. Gugercin, and K. Willcox. A survey of projection-based model reduction methods for parametric dynamical systems. *SIAM Review*, 57(4):483–531, 2015.
- J. Bradbury, R. Frostig, P. Hawkins, M. J. Johnson, C. Leary, D. Maclaurin, G. Necula, A. Paszke, J. VanderPlas, S. Wanderman-Milne, and Q. Zhang. JAX: composable transformations of Python+NumPy programs, 2018.
- K. Carlberg. Adaptive h -refinement for reduced-order models. *International Journal for Numerical Methods in Engineering*, 102(5):1192–1210, 2015.
- K. Carlberg, C. Bou-Mosleh, and C. Farhat. Efficient non-linear model reduction via a least-squares Petrov–Galerkin projection and compressive tensor approximations. *International Journal for Numerical Methods in Engineering*, 86(2):155–181, 2011.
- A. Cohen and R. DeVore. Approximation of high-dimensional parametric PDEs. *Acta Numerica*, 24:1–159, 2015.
- R. Geelen, S. Wright, and K. Willcox. Operator inference for non-intrusive model reduction with quadratic manifolds. *Computer Methods in Applied Mechanics and Engineering*, 403:115717, 2022.
- G. H. Golub and V. Pereyra. The differentiation of pseudo-inverses and nonlinear least squares problems whose variables separate. *SIAM Journal on Numerical Analysis*, 10(2):413–432, 1973.
- C. Greif and K. Urban. Decay of the Kolmogorov N -width for wave problems. *Applied Mathematics Letters*, 96:216–222, 2019.
- J. A. Hernández, M. A. Caicedo, and A. Ferrer. Dimensional hyper-reduction of nonlinear finite element models via empirical cubature. *Computer Methods in Applied Mechanics and Engineering*, 313:687–722, 2017.
- M. R. Hestenes and E. Stiefel. Methods of conjugate gradients for solving linear systems. *Journal of Research of the National Bureau of Standards*, 49(6):409–436, 1952.
- Y. Kim, Y. Choi, D. Widemann, and T. Zohdi. A fast and accurate physics-informed neural network reduced order model with shallow masked autoencoder. *Journal of Computational Physics*, 451:110841, 2022.
- M. Kim, T. Wen, K. Lee, and Y. Choi. Physics-informed reduced order model with conditional neural fields. *arXiv preprint arXiv:2412.05233*, 2024.
- N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural operator: learning maps between function spaces. *Journal of Machine Learning Research*, 24(89):1–97, 2023.
- C. L. Lawson and R. J. Hanson. *Solving Least Squares Problems*. Prentice-Hall, Englewood Cliffs, NJ, 1974.
- K. Lee and K. T. Carlberg. Model reduction of dynamical systems on nonlinear manifolds using deep convolutional autoencoders. *Journal of Computational Physics*, 404:108973, 2020.
- Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.
- L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.
- D. W. Marquardt. An algorithm for least-squares estimation of nonlinear parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963.
- B. Peherstorfer and K. Willcox. Online adaptive model reduction for nonlinear systems via low-rank updates. *SIAM Journal on Scientific Computing*, 37(4):A2123–A2150, 2015.

- O. Ronneberger, P. Fischer, and T. Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.
- L. Sirovich. Turbulence and the dynamics of coherent structures, Parts I–III. *Quarterly of Applied Mathematics*, 45(3):561–590, 1987.
- R. Ștefănescu and A. Sandu. Efficient approximation of sparse Jacobians for time-implicit reduced order models. *arXiv preprint arXiv:1402.2018*, 2014.
- M. Takamoto, T. Praditia, R. Leiteritz, D. MacKinlay, F. Alesiani, D. Pflüger, and M. Niepert. PDEBench: An extensive benchmark for scientific machine learning. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 35, 2022.
- M. Tancik, P. P. Srinivasan, B. Mildenhall, S. Fridovich-Keil, N. Raghavan, U. Singhal, R. Ramamoorthi, J. T. Barron, and R. Ng. Fourier features let networks learn high frequency functions in low dimensional domains. *Advances in Neural Information Processing Systems*, 33, 2020.
- J. Weder, P. Schwerdtner, and B. Peherstorfer. Online-efficient model reduction on quadratic manifolds with Neural Galerkin schemes. *arXiv preprint arXiv:2412.17695*, 2024.
- H. Wu, H. Luo, H. Wang, J. Wang, and M. Long. Transolver: A fast transformer solver for PDEs on general geometries. In *International Conference on Machine Learning*, 2024.
- M. Yano and A. T. Patera. An LP empirical quadrature procedure for reduced basis treatment of parametrized nonlinear PDEs. *Computer Methods in Applied Mechanics and Engineering*, 344:1104–1123, 2019.

A METHOD DETAILS

The generic overdetermined problem (3) is instantiated differently per PDE. Poisson, heat and Burgers below are fully discrete weak least-squares problems; the wave arm (§A.4) is an explicit second-order latent integration and is not an instance of (3). Formulas are for the scalar two-dimensional Dirichlet case; three-dimensional and periodic-vector scope is stated in Appendix C.1.

A.1 ELLIPTIC INSTANCE: POISSON

For $Au = f$ we take $\Lambda_\star = \Lambda$, so the weak residual is

$$r_w(z, y) = \Lambda^{-1}P(Au - f) = B_0(h_\theta(z) + C_q y) - b_0, \quad B_0 = PG, \quad b_0 = \Lambda^{-1}Pf. \quad (5)$$

Because the retained tests are orthonormal and $Au^\star = f$, this residual is exactly the tested error $P(u - u^\star)$, so the minimised objective is the squared discrete L^2 error of the reduced state in the retained modes; this holds only for the modes P retains.

Analytically eliminated corrections. Let $B_0 C_q = QR$ be a thin QR factorisation. Q does not depend on z , so the inner minimisation over y has the closed form $\mathcal{R}y = Q^\top(b_0 - B_0 h_\theta(z))$ and the minimised value is

$$\min_y \|B_0(h_\theta(z) + C_q y) - b_0\| = \|B_\perp h_\theta(z) - b_\perp\|, \quad B_\perp = (I - QQ^\top)B_0, \quad b_\perp = (I - QQ^\top)b_0. \quad (6)$$

The LM iteration runs on z alone with the deflated operator, and y is recovered by one triangular solve. Because the projector is constant the elimination introduces no approximation and $B_\perp D h_\theta$ is the exact Jacobian of the eliminated residual; the only dropped derivative anywhere in the elliptic solve is the head curvature. At $q = R$ the deflated operator is zero, the latent problem is degenerate by construction, and the model is the linear least-squares solve $\min_y \|B_0 y - b_0\|$; the timed top rung is that QR solve, not an LM iteration on a zero Jacobian. We report the reduced stationarity of the deflated problem, the augmented stationarity of the full problem, the backward error of the triangular recovery and the numerical rank of $B_0 C_q$.

A.2 PARABOLIC INSTANCE: HEAT

The semi-discrete heat equation $\dot{u} = -\kappa Au$ is advanced with Crank–Nicolson, $(I + \frac{\Delta t \kappa}{2} A)u^{n+1} = (I - \frac{\Delta t \kappa}{2} A)u^n$, and the reduced step substitutes the manifold into the fully discrete equation *before* projecting. With $u^n = u(z_n)$, projecting with P and row-scaling by $\Lambda_\star = I + \frac{\Delta t \kappa}{2} \Lambda$ gives

$$r_{w,n}(z) = B_0 h_\theta(z) - D B_0 h_\theta(z_n), \quad D = \text{diag}\left(\frac{1 - \frac{\Delta t \kappa}{2} \lambda_i}{1 + \frac{\Delta t \kappa}{2} \lambda_i}\right), \quad (7)$$

and $z_{n+1} = \arg \min_z \|r_{w,n}(z)\|_2$, warm-started at z_n . Two points are structural. The right-hand side is built from the *decoded current state* at every step: carrying the quantity the previous step already made small freezes the recursion and reproduces a one-step solution to round-off. And z_{n+1} enters through h_θ , so the step is a nonlinear least-squares problem, not a linear solve. With corrections, $c_n = h_\theta(z_n) + C_q y_n$ replaces $h_\theta(z_n)$:

$$(z_{n+1}, y_{n+1}) = \arg \min_{z,y} \|B_0(h_\theta(z) + C_q y) - D B_0 c_n\|_2, \quad (8)$$

where y is eliminated as in (6) and the *complete* previous coefficients c_n , corrections included, are carried into the next step. At $q = R$ the head is redundant and the step is the linear least-squares recurrence

$$c_{n+1} = B_0^+ D B_0 c_n, \quad (9)$$

with B_0^+ the pseudo-inverse ($M \geq R$; the numerical rank of B_0 is checked) and c_0 fitted to the supplied initial field. This endpoint is the weak Crank–Nicolson propagation of the bank coefficients, with no head and no iteration. It is the “top rung” of the heat ladder and the linear-bank baseline of the text. The deployed path factors the damped normal matrix by Cholesky; a variant assembles the Gram matrix $S = B_0^\top B_0$ offline and forms $H = D h_\theta^\top S D h_\theta$ directly, which is exact and is reported as an algebraic ablation of the same step.

A.3 NONLINEAR ADVECTION: BURGERS

The full-order problem is $u_t + u(u_x + u_y) = \nu \Delta u$ with the non-conservative sign-upwind stencil

$$N(u)_{ij} = u_{ij} (\delta_x u + \delta_y u)_{ij}, \quad (\delta_x u)_{ij} = N \cdot \begin{cases} u_{ij} - u_{i-1,j}, & u_{ij} > 0, \\ u_{i+1,j} - u_{ij}, & u_{ij} \leq 0, \end{cases} \quad (10)$$

δ_y analogous and switching on the same centre value, ghost zeros on all four walls, and backward Euler in time, $r_n(u) = u - u^n + \Delta t (N(u) - \nu \Delta_h u)$. Projecting and row scaling by $\Lambda_\star = I + \Delta t \nu \Lambda$ gives, with $c = h_\theta(z) + C_q y$,

$$r_{w,n} = (I + \Delta t \nu \Lambda)^{-1} [B_0 c - B_0 c_n + \Delta t (\hat{N}(c) + \nu \Lambda B_0 c)], \quad (11)$$

where the diffusion term is exact and the row scaling is the diagonal of the linearised implicit operator in the modal basis, the modal form of the Helmholtz preconditioner the full-order Newton solver uses. Only $\hat{N}(c) \approx PN(Gc)$ requires a choice: dense evaluation at every interior node, the sampled rule $\hat{N}(c) = \sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(Gc)$ with the sign switch retained at the retained nodes, or the precomputed quadratic form $\frac{1}{2} c^\top Q c$ with Q the symmetrised tensor $T_{iab} = \sum_x P_{ix} G_{xa} (DG)_{xb}$, which is an identity only where $u > 0$ at every stencil node of every trial state the solver visits. The last is an algebraic route we verified in-job (residual and Jacobian parity below 10^{-11} relative on sign-satisfying states) and do not time in this paper. Time stepping uses a fixed substep count per output interval; each step is warm-started at the previous code, and the linear extrapolation is used instead when, and only when, its residual norm is smaller; the 4096^2 rows also try a quadratic predictor under the same rule. In every Burgers arm above 1024^2 except the 2048^2 development accurate arm, the damping λ is carried from one step to the next rather than reset to λ_0 .

Block-damped step. With $J = [J_z \ J_y]$ the Jacobian of (11) in (z, y) , obtained in one forward-mode pass, each iteration solves

$$(J^\top J + \lambda D_z + \varepsilon D_y) \begin{bmatrix} \delta z \\ \delta y \end{bmatrix} = -J^\top r_{w,n}, \quad \|\delta z\| \leq \Delta, \quad (12)$$

where $D_z = \text{diag}(\text{diag}(J_z^\top J_z), 0)$ carries the Levenberg–Marquardt damping on the latent block only, $D_y = \text{diag}(0, I)$ with a fixed small ridge ε , and Δ is the $q = 0$ trust radius. The correction step is therefore an undamped Gauss–Newton step on the current linearisation, not throttled by a radius calibrated for the latent code. The step is accepted only if the residual decreases; otherwise λ grows as in §A.6. Table D.3 compares it with joint damping and with variable projection.

A.4 SECOND-ORDER INSTANCE: REFLECTIVE WAVES

For $u_{tt} = c^2 \Delta u$ the bank is mass-orthonormal and the R weak tests are the bank itself, so projection is Galerkin with stiffness $K = G^\top M_h A G$, M_h the diagonal mass matrix. Writing the coefficients as $a(z, y) = h_\theta(z) + C_q y$ and differentiating twice in time, $\ddot{a} = Dh_\theta \ddot{z} + h''_\theta(z)[\dot{z}, \dot{z}] + C_q \ddot{y}$, the projected equation $\ddot{a} = -c^2 K a$ becomes a linear least-squares problem for the joint acceleration,

$$[Dh_\theta(z) \ C_q] \begin{pmatrix} \ddot{z} \\ \ddot{y} \end{pmatrix} = -(c^2 K a + h''_\theta(z)[\dot{z}, \dot{z}]), \quad (13)$$

whose curvature term $h''_\theta[\dot{z}, \dot{z}]$ is a forward-over-forward derivative of the head. The first-order system in $(z, y, \dot{z}, \dot{y})$ is advanced with classical RK4 at a fixed step, one solve of (13) per stage; there is no implicit residual minimisation and no stationarity exit. Initial coordinates and velocities are obtained from the supplied displacement and velocity. At $q = R$ the head is dropped and $\ddot{a} = -c^2 K a$ is propagated exactly through the eigendecomposition $K = V \Lambda V^\top$, $a(t) = V(\cos(\omega t) \hat{a}_0 + \omega^{-1} \sin(\omega t) \hat{b}_0)$, $\omega = c\sqrt{\Lambda}$. The iterative full-order comparator uses implicit midpoint with CG. The error reported in Table 3 is the energy-state error over displacement and velocity.

A.5 EMPIRICAL QUADRATURE: THE FIT AND THE COUNTS

A fitted rule with support $\mathcal{S}$ and weights w is scored by the held-out relative error of the projected advection term,

$$\rho(u) = \frac{\|\sum_{i \in \mathcal{S}} w_i P_{:,i} N_i(u) - PN(u)\|_2}{\|PN(u)\|_2}, \quad \rho_{\max} \leq 0.116 \text{ (primary bar)}, \quad \rho_{\max} \leq 0.06 \text{ (tight)}, \quad (14)$$

over the held-out reachable states described below. The primary bar is the held-out ρ of the original model’s $q = 0$ rule at the state that carries its largest first-interval error, measured in an earlier diagnostic study and fixed before any certification job ran; the tight bar was fixed before the EQ-ladder job of Table D.2. A candidate node set $\mathcal{C}$ is drawn once with a fixed seed. For each of n_{fit} stored codes $c^{(\ell)}$ we evaluate the decoded state, form the per-node contributions $N_i(Gc^{(\ell)})$ and the exact full-grid target $PN(Gc^{(\ell)})$, and solve

$$\min_{w \geq 0} \|\mathcal{G} w - \beta\|_2, \quad \mathcal{G}_{(\ell,i),c} = P_{i,c} N_c(Gc^{(\ell)}), \quad \beta_{(\ell,i)} = [PN(Gc^{(\ell)})]_i, \quad (15)$$

rows normalised by their Euclidean norms, support grown greedily by the Lawson–Hanson criterion with an exact non-negative refit after each addition and a hard cap of m nodes, padded to the requested count if the support saturates. Three counts are separated: the requested node count m , the achieved support, and the number of fit states n_{fit} . In the certification study the fit states are reachable states (states of the dense solver’s own converged per-step trajectory on fit trajectories), the held-out states are 512 reachable states per rung from certification trajectories disjoint from both the fit and the evaluation cases, and ρ of (14) is evaluated on those. A deployed rule is a stored triple: node indices, weights, and the cached $m \times 5 \times R$ bank block.

A.6 SOLVER CONSTANTS AND EXIT CODES

Damping starts at $\lambda_0 = 10^{-6}$ (Poisson, Burgers) or 10^{-4} (heat); $\lambda \leftarrow \max(\lambda/3, 10^{-12})$ on acceptance and $\min(10\lambda, \lambda_{\max})$ on rejection; the trust radius is taken from the spread of the training codes. Exit codes differ per problem and are reported per problem. Burgers: 4 stationarity, 1 residual, 2 tiny step, 3 rejected at $\lambda \geq 10^{14}$, 0 budget. Poisson: 6 stationarity, 2 relative residual, 1 stall, 3 damping limit, 5 non-finite initial value, 0 budget. Heat: 1 stationarity, 2 tiny step, 3 damping limit, 4 non-finite, 0 budget. The heat criterion divides by $\|J\|_F$ only, applied to a residual already normalised by the target norm; Poisson and Burgers use (4) directly. An initial fit that meets its stopping rule is accepted as the starting state of the time loop; this acceptance rule was fixed before the evaluation jobs ran, and complete exit flags remain in the source records.

B ARCHITECTURE AND ONLINE DATA FLOW

Figure B.1 shows the offline preparation and the online solve of every NM-ROM query in the paper.

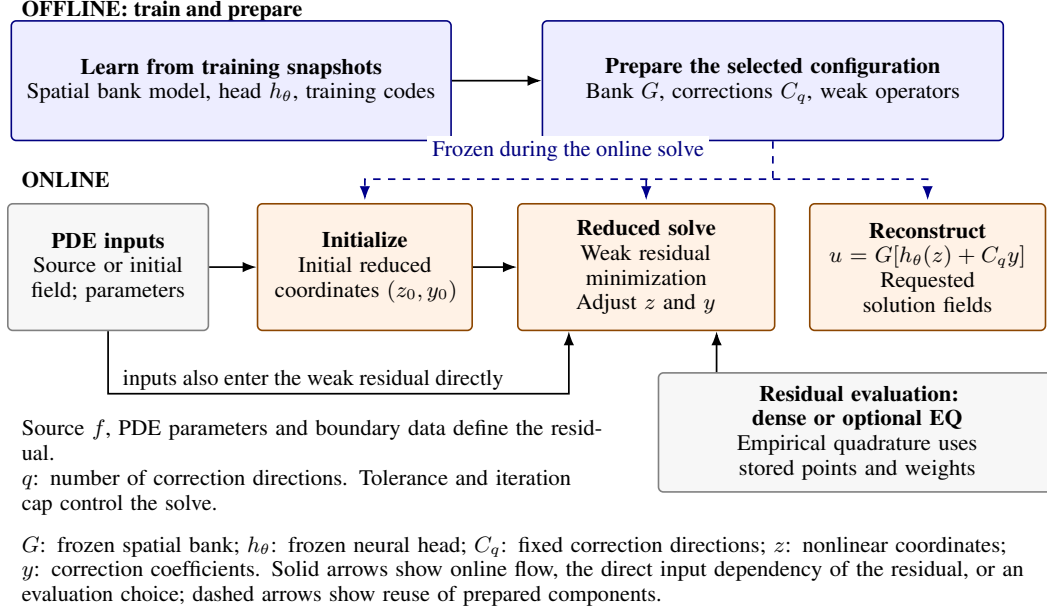


Figure B.1: NM-ROM from training to prediction. Blue components are prepared before the query and remain frozen; orange boxes compute the online solution. The inputs initialize reduced coordinates, which are adjusted to minimize the weak PDE residual before reconstructing the requested fields. Initialization and the reduced solver are PDE-specific; linear-PDE corrections can be eliminated analytically. Time-dependent problems repeat the reduced step, with reconstruction at requested output times. Empirical quadrature (EQ) is an optional residual evaluation, used for two-dimensional Burgers; the three-dimensional and wave solves do not use it. Correction rank changes the representation, whereas EQ changes residual evaluation. Baselines are evaluated independently and are not stages of this pipeline.

C EXPERIMENTAL CONFIGURATION AND REPRODUCIBILITY

Tables C.2 and C.3 identify the two-dimensional studies and Table C.11 the three-dimensional ones. Within each study, training precedes evaluation and the selected bank, head and correction directions remain frozen. Deployment changes reduced coordinates, not network weights. Recorded training configurations and available offline costs are retained with the source evidence; unrecorded training costs are not inferred. The wide heat bank of Table 1 is a random-Fourier-feature uncoordinated network (32 features, scale 1.5, width 256, $R = 128$) trained for 80000 steps (learning rate 0.001, 64 states $\times$ 2048 points per batch, gradient clip 1, exact coefficient refit every 10000 steps), then a head of width 256 with $k = 8$ for 80000 steps (learning rate 0.0005, 128 states per batch), on 512 training draws at 64^2 with 16 validation draws; its validation projection floor is 0.11 %.

Full-order and solver details. The Burgers FOM is Newton’s method with BiCGStab inner solves (at most 200 iterations, Jacobian–vector products by forward-mode differentiation) preconditioned by the exact inverse of the implicit Helmholtz operator $(I + \Delta t \nu \lambda)^{-1}$ (advection dropped); Newton stops when the residual norm falls below the nonlinear tolerance times the previous state’s norm, or after 20 iterations, and always takes the full step (no line search or damping). Each named setting fixes Δt , the nonlinear and the linear tolerance. On the L-shape the vanishing factor is $4 \min(x, y, d_{\text{right}}, d_{\text{top}}, d_{\text{corner}})$ with the distances to the outer right and top edges and to the re-entrant corner, and the test space is the M lowest eigenvectors of the five-point operator on the masked grid, computed by shift-invert Lanczos and scaled as $P = \Lambda^{-1} \Phi^\top$ ($M = 257$; $M = 1024$ for the $q = R$ solve). The quadratic predictor of the 4096^2 Burgers rows forms the linear extrapolation $2w_n - w_{n-1}$ and, from the third step on, the quadratic one $3w_n - 3w_{n-1} + w_{n-2}$, evaluates the

Table C.1: Training configurations of the frozen checkpoints, transcribed from the committed configuration and code files of each study (paths and SHA256 in evidence/training-configs-2026-09-21).

Model	Stage	Network	Optimiser, learning rate	Steps	Batch	Training data
Burgers 2D ($k = 16, R = 512$)	bank	128 Fourier features (scale 4), width 1024, 2 layers	AdamW, wd 10^{-5} , warm-up + cosine $10^{-3} \rightarrow 10^{-5}$	300000	all states $\times$ 4096 points	16384 states of 576 trajectories
	head	width 512, 2 layers	Adam, warm-up + cosine $10^{-3} \rightarrow 10^{-5}$	200000	4096 states	131072 states of 4608 trajectories
Poisson 2D ($k = 32, R = 512$)	bank	64 Fourier features (scale 4), width R , 2 layers	Adam, four phases at 10^{-3} , 3×10^{-4} , 3×10^{-4} , 10^{-4}	100000, 40000, 30000, 30000	64 sources $\times$ 4096 points	2611 fit / 461 validation sources
	head	width 128, 2 layers	Adam, 10^{-3}	150000 (full batch)	all	as bank
Poisson 2D ($k = 16, R = 128$)	bank + head	64 Fourier features, g and h width 128, 2 layers	Adam; staged from an $R = 64$ model (bank, head, joint at 3×10^{-4} , 3×10^{-4} , 10^{-4}) as Poisson $k = 32$	40000, 30000, 20000 after widening	64 sources $\times$ 4096 points	512 training sources
L-shape ($k = 16, R = 512$)	bank / head	64 Fourier features (scale 4), 2 layers; head width 128, 2 layers	Adam; cosine (floor 0.03); 10^{-3} bank, 5×10^{-4} head	100000, 40000, 30000, 30000; head 150000	64 sources $\times$ 4096 points	2611 fit / 461 validation sources
Poisson 3D ($k = 16, R = 128$)	bank / head	64 Fourier features (scale 1.5), widths 256; head width 256 + skip	Adam, cosine (floor 0.03); 10^{-3} bank, 5×10^{-4} head; clip 1, refit every 10000	150000 / 100000	64 states $\times$ 2048 points / 128 states	512 training / 16 validation fields
Heat 3D ($k = 32, R = 128$)	bank / head	32 Fourier features (scale 1.5); head width 256 + skip	Adam, cosine (floor 0.03); 10^{-3} bank, 5×10^{-4} head; clip 1, refit every 10000	150000 / 150000	64 states $\times$ 2048 points / 128 states	512 training / 16 validation trajectories

residual norm of the current code and both candidates in one batched call and starts the solve from the smallest.

Table C.2: Problem specification. Cohort and reduced sizes are read from the run configurations where recorded. The Burgers sealed cohort has been opened and is reported in Table D.1; other rows describe their recorded development and validation cohorts.

PDE	equation, domain, boundary	meshes	time stepping	reduced sizes	reference	cohorts
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 – 4096^2	$\Delta t = 0.005$, backward Euler, sign-upwind	$k = 16, R = 512$	refined 4096^2 , $\Delta t = 0.00015625$	6 development cases; 32 validation; 64 held-out at 2048^2 , 4096^2 ; sealed cohort opened once
Poisson 2D	$-\Delta u = f$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	256^2 – 4096^2	none (elliptic)	$k = 16, R = 128$ (original); $k = 32, R = 512$	exact discrete solution; 2048^2 refinement	12 development sources
Heat 2D	$u_t = \kappa \Delta u$, $(0, 1)^2$	64^2 – 1024^2	Crank–Nicolson	$k = 8, R = 32$	refined-grid reference ($1024^2/2048^2$ pair); error includes discretisation	12 development cases, 3 repetitions; measured in job 3529772; checkpoint lineage: earlier cell, job 3511417
Heat 2D, wide bank	$u_t = \kappa \Delta u$, $(0, 1)^2$	1024^2 – 4096^2	Crank–Nicolson; batched propagator fit	$k = 8, R = 128$	exact same-grid propagation	12 development cases; 16 sealed held-out cases, opened once
Wave 2D (reflective)	$u_{tt} = c^2 \Delta u$, $(0, 1)^2$, $u _{\partial\Omega} = 0$	64^2 , 256^2 , 1024^2	RK4 on the manifold; exact modal propagation for the bank	$k = 32, R = 64$	exact modal solution	8 development cases
Poisson, L-shape	$-\Delta u = f$, $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	256^2 – 2048^2	none	$k \in \{16, 32\}$, $R \in \{256, 512, 514\}$	converged discrete solution	3072 / 256 / 32 sources (train / selection / development)

Table C.3: Sampled problem families, transcribed from the generator sources named in the last column (code constants, not run outputs).

PDE	equation and boundary	sampled family (transcribed from the generator source)	source
Burgers 2D	$u_t + u(u_x + u_y) = \nu \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$u_0 = a \exp(- x - c ^2 / 2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim U(0.05, 0.20)$, $a \sim U(0.5, 2)$; $\nu \sim \log U(0.01, 0.1)$; outputs $t \in \{0.05, \dots, 0.25\}$	experiments/ mr-burgers2d/ engines.py: params_draw
Poisson 2D	$-\Delta u = f$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	$f = a \exp(- x - c ^2 / 2w^2)$, $c_i \sim U(0.15, 0.85)$, $w \sim \log U(0.02, 0.1)$, $a \sim U(0.5, 2)$	multistage- precision/ms_ parametric.py: sample_params
Poisson, L-shape	same source family on $(0, 1)^2 \setminus [\frac{1}{2}, 1)^2$	as above; sources centred in the removed quadrant rejected	experiments/ lshape
Heat 2D	$u_t = \kappa \Delta u$ on $(0, 1)^2$, $\kappa = 0.02$ fixed	polynomial-boundary Gaussian family (single_bc_poly_gaussian_v1); Crank–Nicolson $\Delta t = 0.025$	checkpoint_expanded_ seed790715 (lineage: linear-bank cell, job 3511417); printed CG measurements: job 3529772
Wave 2D (reflective)	$u_{tt} = c^2 \Delta u$ on $(0, 1)^2$, $u = 0$ on $\partial\Omega$	compact bump $\times$ Gaussian: half-widths $s_i \sim U(0.36, 0.42)$, centre $c_i \sim U(s_i + 0.025, 1 - s_i - 0.025)$, amplitude $\sim U(0.7, 1.3)$, $\sigma_i \sim U(0.12, 0.16)$, advective velocity $v_i \sim U(-0.5, 0.5)$ (zero every fourth case); speed $c \sim U(0.85, 1.15)$	experiments/ multiresolution- wave/audit_ dynamics.py: parameter_rows

C.1 HEADLINE SETTINGS, TIMES AND TIMING PROTOCOL

Table C.4 gives the settings and measured times behind every row of Table 1. Every cost and error come from the same invocation. Timings use GPU warm-up and synchronisation, double precision and the highest matrix-multiplication precision; all repetitions, including outliers, enter the median, and no time is borrowed from another allocation. Requested full fields and initialisation are charged; host transfer is included only where the table says complete query. For the held-out Burgers rows, an independent restricted-grid recomputation of the errors disagreed with the full-grid value by more than 5 % on 5/384 (2048²) and 18/1152 (4096²) case-arm rows; per-arm cohort worsts agree within 1.2 % and the full-grid recomputation is exact. For Burgers at 4096², copying the six double-precision output fields to the host adds about 246 ms to every arm. The Poisson and three-dimensional heat CG comparators are unpreconditioned; heat CG is warm-started from the previous time level. The Heat2D rows use checkpoint `expanded_seed790715`; Table C.2 separates that checkpoint’s lineage from the job that produced the printed measurements.

Table C.4: Supporting data for Table 1: the two settings of each frozen model, median times (ms), timing scope and evidence status (allocations: Appendix C). “EQ” and “dense” name the Burgers residual evaluation; “single” marks models measured at one setting. Where a run recorded both timing scopes, the speedups in the scope not used by Table 1 are listed; a series never mixes scopes. For heat that column gives the speedups against the tighter CN-CG setting (rtol 10⁻⁶). * not in Table 1: a re-measurement of the row above it, or a development-source run at a mesh whose held-out row is in Table 1.

Problem	Mesh	Accurate	Fast	Accurate ms	Fast ms	FOM ms	Timing	Other scope or FOM: acc. / fast	Status
Poisson 2D	256 ²	$q = 256$	$q = 0$	7.12	6.70	22.29	GPU query	—	development
Poisson 2D	1024 ²	$q = 256$	$q = 0$	4.32	3.94	60.02	GPU query	—	development
Poisson 2D	2048 ²	$q = 256$	$q = 0$	5.47	5.35	397.80	GPU query	28.7 × / 28.8 × (complete query)	development
Poisson 2D	4096 ²	$q = 256$	$q = 0$	16.92	16.74	2472.81	GPU query	41.1 × / 41.7 × (complete query)	development
Poisson 2D*	4096 ²	$q = 256$	$q = 0$	17.17	17.01	2465.60	GPU query	37.6 × / 37.9 × (complete query)	development
Poisson, L-shape 2D	256 ²	$q = 64$	$q = 0$	3.03	2.88	11.64	complete query	—	development
Poisson, L-shape 2D	512 ²	$q = 64$	$q = 0$	4.80	4.72	29.13	complete query	—	development
Poisson, L-shape 2D	1024 ²	$q = 128$	$q = 0$	5.81	5.67	56.72	complete query	16.0 × / 16.5 × (GPU query)	development
Poisson, L-shape 2D	2048 ²	$q = 128$	$q = 0$	18.75	18.66	362.30	complete query	33.9 × / 34.3 × (GPU query)	development
Heat 2D	64 ²	—	single	—	11.23	2.50	GPU query	—	development
Heat 2D	256 ²	—	single	—	11.34	6.94	GPU query	—	development
Heat 2D	1024 ²	—	single	—	12.21	59.18	GPU query	—	development
Heat (wide bank) 2D	1024 ²	$q = 32$	$q = 0$	27.33	27.77	43.51	GPU query	4.70 × / 4.62 × (vs. CN-CG rtol 10 ⁻⁶)	sealed held-out
Heat (wide bank) 2D	2048 ²	$q = 32$	$q = 0$	27.97	28.42	272.55	GPU query	28.8 × / 28.3 × (vs. CN-CG rtol 10 ⁻⁶)	sealed held-out
Heat (wide bank) 2D	4096 ²	$q = 32$	$q = 0$	31.68	32.19	1139.83	GPU query	149 × / 146 × (vs. CN-CG rtol 10 ⁻⁶)	sealed held-out
Heat (wide bank, batched fit) 2D	1024 ²	$q = 32$	$q = 0$	3.13	3.36	43.51	GPU query	41.0 × / 38.3 × (vs. CN-CG rtol 10 ⁻⁶)	sealed held-out
Heat (wide bank, batched fit) 2D	2048 ²	$q = 32$	$q = 0$	4.54	4.78	272.55	GPU query	177 × / 169 × (vs. CN-CG rtol 10 ⁻⁶)	sealed held-out
Heat (wide bank, batched fit) 2D	4096 ²	$q = 32$	$q = 0$	11.03	11.33	1139.83	GPU query	428 × / 416 × (vs. CN-CG rtol 10 ⁻⁶)	sealed held-out
Burgers 2D	256 ²	$q = 256, M = 1088, \text{EQ } m = 2560$	$q = 0, M = 64, \text{EQ } m = 1024$	746.02	40.36	31.79	GPU query	—	development
Burgers 2D	512 ²	$q = 256, M = 1088, \text{EQ } m = 2438$	$q = 0, M = 64, \text{EQ } m = 922$	783.33	40.49	52.92	GPU query	—	development
Burgers 2D	1024 ²	$q = 256, M = 1088, \text{dense}$	$q = 0, M = 64, \text{EQ } m = 934$	2203.85	40.27	81.31	GPU query	—	development
Burgers 2D	2048 ²	$q = 256, M = 544, \text{EQ lattice } m = 3969$	$q = 0, M = 64, \text{EQ } m = 1024$	124.15	29.48	122.76	GPU query	0.99 × / 2.01 × (complete query); 6.07 × / 25.5 × (vs. tight Newton)	development
Burgers 2D	4096 ²	$q = 256, M = 1088, \text{EQ lattice } m = 3969$	$q = 0, M = 64, \text{EQ } m = 1024$	107.51	40.73	523.85	GPU query	2.18 × / 2.68 × (complete query); 30.4 × / 80.3 × (vs. tight Newton)	development
Burgers (held-out cases) 2D	2048 ²	$q = 256, M = 1088, \text{EQ lattice } m = 3969$	$q = 0, M = 64, \text{EQ } m = 1024$	114.91	29.19	164.43	GPU query	1.28 × / 2.49 × (complete query); 6.48 × / 25.5 × (vs. tight Newton)	held-out
Burgers (held-out cases) 2D	4096 ²	$q = 256, M = 1088, \text{EQ lattice } m = 3969$	$q = 0, M = 64, \text{EQ } m = 1024$	99.96	40.60	537.39	GPU query	2.28 × / 2.74 × (complete query); 32.3 × / 79.6 × (vs. tight Newton)	held-out
Burgers (earlier model) 2D	1024 ²	—	single	—	41.68	68.04	GPU query	—	development
Burgers (earlier model) 2D	1024 ²	—	single	—	41.68	605.75	GPU query	—	development
Poisson 3D	32 ³	$q = 96$	$q = 0$	2.49	2.47	GPU query	—	—	accepted final
Poisson 3D	64 ³	$q = 96$	$q = 0$	3.36	3.22	4.46	GPU query	—	accepted final
Poisson (dev. sources) 3D*	64 ³	$q = 96$	$q = 0$	1.43	1.38	2.88	GPU query	1.57 × / 1.56 × (complete query)	development
Poisson (dev. sources) 3D	128 ³	$q = 96$	$q = 0$	1.86	1.80	12.55	GPU query	2.70 × / 2.66 × (complete query)	development
Poisson (dev. sources) 3D*	128 ³	$q = 96$	$q = 0$	1.90	1.76	12.40	GPU query	2.65 × / 2.70 × (complete query)	development
Poisson (dev. sources) 3D	256 ³	$q = 96$	$q = 0$	6.08	5.95	140.87	GPU query	4.18 × / 4.21 × (complete query)	development

Table C.5: Heat at high resolution, accuracy: one frozen model per block; worst same-grid relative L^2 error over all output times and over evolved times, accurate / fast setting. CN: reduced Crank–Nicolson steps; batched fit: each output time fitted independently to exactly propagated test moments (linear autonomous problems with eigenfunction tests only). The sealed 2D cohort was opened once and gives every wide-bank row of Table 1; the development row ran on another GPU (A100) and is shown for reference. Wide bank: $R = 128, k = 8$; earlier: the $R = 32$ checkpoint of the Heat rows; 3D: the Heat 3D model of Table 3. The 3D rows miss the 1 % all-times target; 256³ is the first 16 final cases, not a mesh trend.

Model	Mesh	Cohort	Stepping	q acc./fast	Err. all times (%)	Err. evolved (%)
Wide bank	1024 ²	development (12)	CN	32 / 0	0.24 / 0.56	0.17 / 0.46
	1024 ²	sealed (16)	CN	32 / 0	0.49 / 1.36	0.22 / 0.84
	1024 ²	sealed (16)	batched fit	32 / 0	0.49 / 1.36	0.21 / 0.84
	2048 ²	sealed (16)	CN	32 / 0	0.49 / 1.36	0.22 / 0.84
	2048 ²	sealed (16)	batched fit	32 / 0	0.49 / 1.36	0.21 / 0.84
	4096 ²	sealed (16)	CN	32 / 0	0.49 / 1.36	0.22 / 0.84
Earlier ($R=32$)	4096 ²	sealed (16)	batched fit	32 / 0	0.49 / 1.36	0.21 / 0.84
	2048 ²	development (12)	CN	24 / 0	1.68 / 4.56	1.11 / 4.56
3D model	4096 ²	development (12)	CN	24 / 0	1.68 / 4.56	1.11 / 4.56
	128 ³	final (64)	CN	96 / 0	1.93 / 3.18	0.75 / 1.54
	128 ³	final (64)	batched fit	96 / 0	1.93 / 3.18	0.72 / 1.36
	256 ³	final, 1st 16 (16)	CN	96 / 0	1.27 / 2.02	0.55 / 0.91
	256 ³	final, 1st 16 (16)	batched fit	96 / 0	1.27 / 2.02	0.50 / 0.86

Table C.6: Heat at high resolution, timing for the rows of Table C.5: median GPU query times (ms), the FOM chosen by the rule of Table 1 and its speedup, and the speedup against CN-CG at $\Delta t = 0.025$, $\text{rtol } 10^{-6}$.

Model	Mesh	Cohort	Stepping	ms acc./fast	FOM ms	CN-CG FOM	Speedup	vs. 10^{-6}
Wide bank	1024 ²	development (12)	CN	31.2 / 36.9	66.8	$\Delta t=0.05$, $\text{rtol } 10^{-3}$	2.14 / 1.81 $\times$	6.33 / 5.35 $\times$
	1024 ²	sealed (16)	CN	27.3 / 27.8	43.5	$\Delta t=0.05$, $\text{rtol } 10^{-3}$	1.59 / 1.57 $\times$	4.70 / 4.62 $\times$
	1024 ²	sealed (16)	batched fit	3.1 / 3.4	43.5	$\Delta t=0.05$, $\text{rtol } 10^{-3}$	13.9 / 13.0 $\times$	41.0 / 38.3 $\times$
	2048 ²	sealed (16)	CN	28.0 / 28.4	272.5	$\Delta t=0.05$, $\text{rtol } 10^{-3}$	9.74 / 9.59 $\times$	28.8 / 28.3 $\times$
	2048 ²	sealed (16)	batched fit	4.5 / 4.8	272.5	$\Delta t=0.05$, $\text{rtol } 10^{-3}$	60.0 / 57.0 $\times$	177 / 169 $\times$
	4096 ²	sealed (16)	CN	31.7 / 32.2	1139.8	$\Delta t=0.05$, $\text{rtol } 10^{-2}$	36.0 / 35.4 $\times$	149 / 146 $\times$
Earlier ($R=32$)	4096 ²	sealed (16)	batched fit	11.0 / 11.3	1139.8	$\Delta t=0.05$, $\text{rtol } 10^{-2}$	103 / 101 $\times$	428 / 416 $\times$
	2048 ²	development (12)	CN	15.9 / 8.9	173.1	$\Delta t=0.1$, $\text{rtol } 10^{-2}$	10.9 / 19.5 $\times$	51.1 / 91.4 $\times$
	4096 ²	development (12)	CN	18.0 / 10.8	1014.6	$\Delta t=0.1$, $\text{rtol } 10^{-2}$	56.5 / 93.7 $\times$	267 / 442 $\times$
3D model	128 ³	final (64)	CN	35.1 / 16.5	7.0	$\Delta t=0.1$, $\text{rtol } 10^{-2}$	0.20 / 0.43 $\times$	0.83 / 1.78 $\times$
	128 ³	final (64)	batched fit	5.2 / 4.0	7.0	$\Delta t=0.1$, $\text{rtol } 10^{-2}$	1.36 / 1.77 $\times$	5.67 / 7.40 $\times$
	256 ³	final, 1st 16 (16)	CN	37.0 / 21.0	173.3	$\Delta t=0.05$, $\text{rtol } 10^{-4}$	4.69 / 8.27 $\times$	8.96 / 15.8 $\times$
	256 ³	final, 1st 16 (16)	batched fit	9.6 / 8.4	173.3	$\Delta t=0.05$, $\text{rtol } 10^{-4}$	18.1 / 20.5 $\times$	34.7 / 39.3 $\times$

Table C.7: Supporting data for Table 2: every row of that table with its residual path and median GPU query time from the same allocation, plus the exploratory hyper-reduced Kim et al. rows, whose hyper-reduction did not pass its reproduction gate. Our rows use the unoptimised dense reference path (no empirical quadrature, stopping tolerance 10^{-6} , $M = 4(k + q)$ tests); our optimised query is timed only in Table 1, in other jobs, and no ratio across jobs is formed.

Mesh	Method	Residual path	k	Worst (%)	Median (%)	GPU ms (same job)	MB
256 ²	Kim et al. NM-LSPG	dense	8	163.93	43.34	1301	2480
256 ²	Kim et al. NM-LSPG	dense	16	145.40	43.87	2384	2567
256 ²	Kim et al. NM-LSPG	dense	32	127.51	43.61	3846	2742
256 ²	Kim et al. NM-LSPG, data-matched	dense	16	120.33	38.71	3886	2567
256 ²	POD-LSPG	dense	8	55.52	22.72	138	18
256 ²	POD-LSPG	dense	16	41.53	10.45	259	30
256 ²	POD-LSPG	dense	32	24.78	5.65	519	55
256 ²	This work, fast ($q = 0$, $k = 16$)	dense (reference path)	16	6.79	0.66	262	625
256 ²	This work, accurate ($q = 256$, $k = 16$)	dense (reference path)	272	0.88	0.089	3746	1769
256 ²	Kim et al. NM-LSPG-HR (exploratory)	hyper-reduced, not reproduced	8	115.60	51.78	65	2452
256 ²	Kim et al. NM-LSPG-HR (exploratory)	hyper-reduced, not reproduced	16	202.87	52.51	126	2506
256 ²	Kim et al. NM-LSPG-HR (exploratory)	hyper-reduced, not reproduced	32	122.68	41.98	475	2794
512 ²	Kim et al. NM-LSPG	dense	8	175.71	55.48	746	9959
512 ²	Kim et al. NM-LSPG	dense	16	174.99	58.56	1497	10511
512 ²	Kim et al. NM-LSPG	dense	32	252.33	56.72	2059	10879
512 ²	POD-LSPG	dense	8	55.82	24.08	314	71
512 ²	POD-LSPG	dense	16	42.01	13.83	617	122
512 ²	POD-LSPG	dense	32	25.63	7.74	383	222
512 ²	This work, fast ($q = 0$, $k = 16$)	dense (reference path)	16	7.72	0.69	404	2368
512 ²	This work, accurate ($q = 256$, $k = 16$)	dense (reference path)	272	1.05	0.080	6269	6891
512 ²	Kim et al. NM-LSPG-HR (exploratory)	hyper-reduced, not reproduced	8	273.27	53.08	30	9813
512 ²	Kim et al. NM-LSPG-HR (exploratory)	hyper-reduced, not reproduced	16	185.17	58.52	62	10099
512 ²	Kim et al. NM-LSPG-HR (exploratory)	hyper-reduced, not reproduced	32	194.47	56.00	138	10569

Table C.8: Burgers 2D correction settings from one frozen model at 256², test count fixed at $M = 1088$, dense residual, one allocation. Errors are worst evolved same-grid percentages on development cases; times are paired median GPU query times. All rows meet the stopping rule.

Correction rank q	Relative L^2 error (%)	GPU query time (ms)
0	1.2657	848.0
64	1.0593	1359.4
128	0.8711	1886.5
256	0.5194	4377.9

Table C.9: Which knob to turn, measured on Burgers 2D from frozen models. Rank: fixed- M ladder at 256² (Table C.8). EQ: dense time over EQ time at $q = 0$ in one allocation per mesh. Tolerance and cap: 32 validation cases (Table D.4), stopping tolerance $10^{-8} \rightarrow 10^{-3}$ and iteration cap 2. * Jobs 3789570 and 3789572; Table C.10 is another allocation.

Knob	Turn it to	Effect on error	Effect on cost
Correction rank q	0 $\rightarrow$ 256	2.44 $\times$ lower	5.16 $\times$ higher
Empirical quadrature	dense $\rightarrow$ stored rule	Table C.10	4.77 $\times$ lower at 256 ² , 38.53 $\times$ at 1024 ² *
Stopping tolerance	looser	6.712 $\rightarrow$ 6.701 %	35.6 % lower
Iteration cap	2 iterations	fails (90.3 %)	—

Table C.10: Dense and empirical-quadrature (EQ) residual evaluation on Burgers 2D at 256^2 . Error is worst evolved relative L^2 (%); time is median GPU milliseconds from one allocation (job 3780164, the EQ-ladder job of Table D.2; the 256^2 row of Table 1 is job 3789570 and Table C.8 is the fixed- M ladder job, each with its own measured error); the last column is dense time over EQ time. Dashes: no paired dense measurement. Rules at $q = 0, 16, 32$ pass every independent reconstruction; higher-rank rules pass some (Table D.2).

Correction rank q	Dense error (%)	Dense ms	EQ error (%)	EQ ms	Dense / EQ
0	1.8890	292.87	1.8891	59.07	$4.96\times$
16	—	—	1.4270	80.61	—
32	—	—	1.2493	97.70	—
64	1.0843	624.93	1.2275	120.63	$5.18\times$
128	0.8930	1190.49	0.8936	246.87	$4.82\times$
256	0.5194	3915.14	0.5389	722.21	$5.42\times$

Table C.11: Three-dimensional configurations, read from the run records. Domains are the unit cube with homogeneous Dirichlet data, except Navier–Stokes, which is periodic. Burgers and Navier–Stokes appear in Table 3.

Problem	Mesh	Cases	Correction ranks	Named FOM	Residual
Burgers 3D	32^3	32 final	0, 192	Newton–BiCGStab, $\Delta t = 0.01$	dense
Poisson 3D	$32^3, 64^3$	64 final	0, 32, 96 ($k = 16$)	CG, rtol 10^{-2} , no preconditioner	dense
Heat 3D	$32^3, 64^3$	64 final	0, 32, 64, 96 ($k = 32$)	CN–CG, $\Delta t = 0.05$, rtol 10^{-4} , warm start	dense
Heat 3D	128^3	64 final (all times)	0, 96 ($k = 32$)	CN–CG, $\Delta t = 0.1$, rtol 10^{-2}	dense
Navier–Stokes 3D	32^3 periodic	32 final	0, 32, 64, 128, 256 ($k = 64$, $R = 1536$, $M = 2048$)	CNAB2, $\Delta t = 0.01$	dense

Three-dimensional Poisson ($-\Delta u = f$, Gaussian sources), heat ($u_t = \kappa \Delta u$, Gaussian initial fields) and scalar Burgers ($u_t + u(u_x + u_y + u_z) = \nu \Delta u$, backward Euler, sign-upwind) are posed on the unit cube with zero Dirichlet walls and extend the operators of Appendix A across three axes. Navier–Stokes uses periodic vector fields, an incompressibility projection and the CNAB2 integrator; its FOM is not a CG solve. All four three-dimensional problems are evaluated on held-out final cohorts whose settings were frozen beforehand.

C.2 SOURCE RECORDS

The accompanying source package retains the full experiment archive, including failed configurations, retractions, training records, checkpoints, case-level fields, timing repetitions and independent audits. The PDF contains the evidence needed for its stated claims rather than every experiment in that archive. Table generators and frozen evidence identify the displayed values through these machine-readable manifests:

- `tables/provenance.json`: original two-dimensional studies.
- `tables/rewrite-provenance.json`: paired CG comparisons.
- `tables/burgers-iterative-provenance.json`: the earlier fine-grid Burgers tight and relaxed controls.
- `tables/headline-provenance.json`: every row of Tables 1, C.4, 3 and C.11 and every point of Figure 1, with the commit and SHA256 of each source record.

Each manifest records source hashes; its corresponding evidence retains solver configuration, checkpoint identity and allocation metadata. Development selection and final evaluation are distinct. Solver exit flags document numerical stopping, not a proof of global optimality.

Allocations. Every ratio pairs times from one Slurm allocation. Tables 1 and C.4: Poisson 2D 3780692, 3783813, 4049279, 4051236, 4071227; Poisson, L-shape 2D 3784663, 3789568, 4057694; Heat 2D 3529772; Heat (wide bank) 2D 4060928; Heat (wide bank, batched fit) 2D 4060928; Burgers 2D 3789570, 3805065, 3789572, 4071616, 4079320; Burgers (held-out cases) 2D 4077566, 4079321; Burgers (earlier model) 2D 3534502; Poisson 3D 4028642; Poisson (dev. sources) 3D 4051032, 4056288. Table 3: Burgers 3D 4021709; Navier–Stokes 3D 4027788; Heat 3D 4033346, 4072943; Wave 2D 3780450. Table C.11: Burgers 3D 4021709; Poisson 3D 4028642; Heat 3D 4033346;

Navier–Stokes 3D 4027788. Table C.5: 4051290, 4054869, 4060928, 4072943. Tables 2 and C.7: 256^2 4095408, 512^2 4107272 (reproduction gate: 4077574).

Training/evaluation separation (Burgers). The 4608 training trajectories of the Burgers checkpoint were regenerated from their seeds and compared with every evaluation cohort (the 32 validation cases of Table 2, the 64 held-out cases, the 6 development cases, the sealed cohort): no cohort shares a parameter vector, or any single parameter value, with the training set (check script and output in evidence/burgers-train-eval-overlap-2026-09-21).

D VALIDATION OF THE CORRECTION AND QUADRATURE STUDIES

Table D.1: The sealed cohort (job 3804465, NVIDIA A100-PCIE-40GB), opened once after every choice was frozen. Top: per rung of the dense $M = 4(k + q)$ ladder, the original model’s sealed worst evolved error and device time, the seed mean $\pm$ sample standard deviation on the sealed and the development cohort, the pre-registered sealed-over-development ratio of seed means (raw and difficulty-normalised) and the converged count over the 4 checkpoints. Bottom: per-checkpoint verdicts. The original model’s $q = 0$ rung is a single sealed case converged to a wrong branch (gradient exit, zero budget exits); the pre-registered ratio criterion fails at $q = 0$ for the original model alone (5.35) and the convergence criterion fails on the second retrained seed at $q = 64$ (3 budget exits). Top-rank sealed errors per checkpoint: original 0.68 %, first retrained seed 0.61 %, second 0.59 %, third 0.66 %.

q	M	original model sealed %	ms	seeds sealed %	seeds development %	sealed / dev	normalised	converged
0	64	10.1120	375.1	2.8023 ± 0.6357	2.0002 ± 0.4986	1.40	1.41	4 of 4
16	128	1.4617	480.8	1.7305 ± 0.1297	1.5988 ± 0.0995	1.08	1.09	4 of 4
32	192	1.4458	540.5	1.5044 ± 0.1422	1.3727 ± 0.1314	1.10	1.14	4 of 4
64	320	1.3252	754.5	1.3528 ± 0.1099	1.2130 ± 0.0580	1.12	1.13	3 of 4
128	576	1.0964	1460.4	1.0461 ± 0.0619	0.9562 ± 0.0264	1.09	1.17	4 of 4
256	1088	0.6789	6725.5	0.6188 ± 0.0345	0.4987 ± 0.1164	1.24	1.20	4 of 4

checkpoint	job	monotone (evolved)	monotone (all-times)	every setting converged	error span	cost span	secondary criterion
original	3804465	yes	yes	yes	$14.89 \times$	$17.93 \times$	yes
first seed	3804465	yes	yes	yes	$3.52 \times$	$13.25 \times$	yes
second seed	3804465	yes	yes	no	$4.81 \times$	$13.17 \times$	no
third seed	3804465	yes	yes	yes	$5.22 \times$	$14.62 \times$	yes

Table D.2: The EQ ladder with the cheapest rule passing the primary bar in its draw per rung, timed in one allocation (job 3780164, A100 80 GB), with its same-job dense twins, and the construction status from the four-draw replication (job 3783811): confirmed = every re-draw passes, marginal = only the listed fraction does. Errors are as measured with these rules (SHA256-identified); they are not properties of the construction above $q = 32$. Ladder rows are read from the study’s final summary record.

q	m	fit states	$\rho_{\max}$ (this draw)	construction status	EQ evolved %	EQ all %	EQ ms	dense evolved %	dense ms	EQ/dense cost
0	1024	64	0.0153	confirmed (3 of 3 re-draws)	1.8891	2.5629	59.1	1.8890	292.9	0.202
16	1024	64	0.0935	confirmed (3 of 3 re-draws)	1.4270	2.4806	80.6	—	—	—
32	1024	42	0.0533	confirmed (2 of 2 re-draws)	1.2493	2.3534	97.7	—	—	—
64	1024	25	0.0531	marginal (2 of 6 draws pass)	1.2275	2.1489	120.6	1.0843	624.9	0.193
128	2048	64	0.0669	marginal (4 of 5 draws pass)	0.8936	1.8116	246.9	0.8930	1190.5	0.207
256	2048	64	0.1074	marginal (1 of 5 draws pass)	0.5389	0.9053	722.2	0.5194	3915.1	0.184

Table D.3: How the corrections are solved on Burgers, one job (job 3734098, NVIDIA A100 80GB PCIe): joint LM on (z, y) , block-damped, and plain variable projection at $q = 64$ and 128. Same error where all converge; the block-damped step is the one kept.

arm	M	worst all-times %	budget exits	all stationary	device ms
$q=64$, joint LM on (z, y)	320	2.1489	6	no	1119.7
$q=64$, block-damped	320	2.1489	0	yes	619.2
$q=64$, plain variable projection	320	2.1489	0	yes	9527.9
$q=128$, plain variable projection	256	1.8116	297	no	47566.9
$q=128$, block-damped	256	1.8116	0	yes	825.5

Reading the validation tables. Correction rank q is the number of added coefficient directions; M is the number of weak test modes and m the quadrature node count. A scheduled ladder changes M with q ; the main fixed-test-space study holds M constant. A sealed cohort is opened only after choices

Table D.4: Solver-side controls on Burgers 2D, 32 validation cases, one allocation: stopping tolerance, EQ node count m and iteration cap, with full-order settings for scale. Error: worst over cases and output times of the L^2 error against the refined reference, relative to the initial-field norm, so the full-order rows show their discretisation error.

setting	worst % (32 validation)	device ms	early-stopped
EQ $m=256$, tol 10^{-6} , cap 180	7.245	51.0	0/96
EQ $m=512$, tol 10^{-8}	6.712	64.7	0/96
EQ $m=512$, tol 10^{-3}	6.701	41.6	0/96
EQ $m=512$, cap 2 (early-stopped)	90.324	28.4	96/96
FOM Newton 10^{-2} , $\Delta t=0.01$	6.807	9.9	0/96
FOM Newton 10^{-2} , $\Delta t=0.005$	35.357	17.4	0/96
FOM Newton 10^{-4} , $\Delta t=0.005$	6.171	22.4	0/96
FOM Newton 10^{-6} , $\Delta t=0.005$	6.172	88.3	0/96

are frozen. Confirmed quadrature passes every independent reconstruction of the rule; marginal means some reconstructions fail. These statuses concern rule construction, not just the error of one selected trajectory. Same-grid error and error against a refined reference measure different quantities.